Applied Ergonomics 45 (2014) 825–832

Contents lists available at ScienceDirect

Applied Ergonomics

journal homepage: www.elsevier.com/locate/apergo

New chording text entry methods combining physical and virtual buttons on a mobile phone

Fong-Gong Wu*, Yu-Chun Huang, Meng-Long Wu

National Cheng Kung University, Department of Industrial Design, Tainan 70101, Taiwan

ARTICLE INFO

Article history:
Received 17 January 2013
Accepted 21 October 2013

Keywords:
Text entry
Chording keyboard
Mobile phones

ABSTRACT

Traditional mobile phones depend on MultiTap, virtual or physical QWERTY keyboard for text entry, and they had some respective drawbacks include low input performance, occupying too large an area, high error rates, lack of feedbacks, etc. Therefore, some researches utilized the characteristics of the chording keyboard to improve input performance. Yet, as the learning efficiency of the chording keyboard is too low, users are not highly willing to learn. In view of that, this study combines the physical and virtual keys, and develops two chording input methods, MagArea and MemoryTap. After three days of learning, the fourteen experiment participants show effectively reduce error rates on MagArea, and they enhance their input speed on MemoryTap. In addition, excellent learning efficiency is found in the two methods, will be more motivated and willing to employ.

© 2013 Elsevier Ltd and The Ergonomics Society. All rights reserved.

1. Introduction

Traditional cellphones mainly depend on MultiTap for text entry, the advantage of which is that it needs far fewer keys than a conventional keyboard. Yet, its input requires multiple or repeated presses, so the key strokes per character (KSPC) are increased (MacKenzie, 2002). Not only is input efficiency reduced, but it is also more likely to cause repetitive strain injuries (RSI) to a hand (Balakrishnan and Yeow, 2008). Consequently, the QWERTY layout is the mainstream mode of text entry widely adopted by newly-emerging smartphones. Whether installed physically or virtually on a cellphone, the standard QWERTY keyboard, which is normally 28 cm in width, is miniaturized on a 6-cm panel. Needless to say, the key size is not in line with most human fingers (Wigdor, 2004). To address this problem, many researchers have developed a large number of chording input devices, but the fact that their learning efficiency are too low has led to severe criticism (Kristensson, 2007; Tinwala and MacKenzie, 2009). With respect to the above, this study endeavored to develop new modes of chording text entry, which combine physical with virtual keys simultaneously pressed, as well as comply with principles of ergonomic operation and cognitive factors. In this way, the proposed text entry methods reduce the problems with input efficiency and operational errors caused by overly crowded and small keys on common cellphones and chording devices.

1.1. Chording text entry on mobile phones

To enter a character or command with a chording keyboard, users must press multiple keys simultaneously. For instance, the Half-QWERTY keyboard (Matias et al., 1993) relies on the space key as its control unit. Though it has only the left side keyboard, the left key is combined with the space key to produce a corresponding character on the right side through mirroring. Thus, the chording keyboard possesses the following features: fewer keys, a smaller size, high portability, operable in a mobile or unstable environments, and prevention of injury caused by repetitive force-applying (Billinghurst et al., 2009). Accordingly, it is suitable for mobile devices.

Twiddler was invented as a one-handed chording device (Lyons et al., 2006). With its keys arranged in alphabetic order, Twiddler enters a letter through the chording text entry of multiple keys. In addition, it can define a wide range of input characters, so there are many combinations that can be defined for the input of multiple characters, such as -ing and -tion (Wigdor, 2004). Although Twiddler can enhance the input efficiency of characters, some studies have shown that its floor entry rate is too low, its learning time too long, and its error rate too high (Kristensson, 2007; Tinwala and MacKenzie, 2009). Furthermore, with too many keys present, its key layout will have various influences on users of different gender, palm size, and finger length (Lyons et al., 2004). As a result, it is somewhat difficult to convert the hand-held Twiddler into a device fully compatible with mobile phones.

Wigdor and Balakrishnan developed two new modes of chording text entry for cellphones, i.e., TiltText and ChordTap (Wigdor

* Corresponding author.
E-mail addresses: fonggong@mail.ncku.edu.tw, wu.oryang@hinet.net (F.-G. Wu).

0003-6870/$ – see front matter © 2013 Elsevier Ltd and The Ergonomics Society. All rights reserved.
http://dx.doi.org/10.1016/j.apergo.2013.10.011

and Balakrishnan, 2003). TiltText combines the MultiTap layout and four directional movements, namely, forward, backward, left, and right movements, to select a particular character. A similar concept is also found in ChordTap (Wigdor, 2004). ChordTap adds three chording keys to the back of the traditional 12-key flip phone, and the three keys are used to control the selection of characters on the front keyboard. This means that users must press the numeric key with one hand, and press one of the chord keys with the other. The above two methods divide the letters into individual groups; then, in the process of typing, a certain group is first selected through one key, and then a member of the group is selected through the corresponding key or the tilt angle. Thus, the problem connected with the high KSPC is effectively relieved. However, it is rather difficult for users to execute such a two-stage selection simultaneously, so input efficiency is still not improved.

1.2. Ergonomic rules and design elements

Text entry is a fairly complex process, involving visual, tactile, motion, memory, learning, and other cognitive functions. Moreover, they belong to different levels of cognitive processing. The process of text entry is subject to the biomechanical constraints of the hand. Whether perceptual and cognitive concepts are suitable will dominantly affect the user's final performance, such as the response time and accuracy rate of typing (Logan, 2003). Therefore, if a new input method is to be designed, ergonomic operation and cognitive compatibility must be emphasized.

According to Fitts' law concerning motion execution (Fitts, 1954), the keyboard layout should be arranged so that a finger travels the minimum distance necessary, allowing text entry efficiency to be improved. Moreover, some studies have shown that if two hands are used simultaneously to do a non-computing job, operational efficiency will be enhanced because of two-handed coordination (Sandnes and Aubert, 2007). Further, users' familiarity with the QWERTY keyboard, procedural memory, which here concerns fingers corresponding to different letters, should be incorporated into ergonomic considerations when typing actions are designed. Consequently, users need not invest too much cognitive effort while typing (McLoone et al., 2003). Realizing that the procedural memory has a subtle effect on users, many researchers have employed the transfer effect of learning to conduct relevant designs or studies (Hwang and Lee, 2005; MacKenzie and Soukoreff, 2002; Matias et al., 1993; McLoone et al., 2003; Sandnes and Aubert, 2007). Contrarily, if the layout design is different from the user's original procedural memory, it will cause interference (McLoone et al., 2003). In addition to enhancing motional performance, researchers have also changed the user interface, singled out the character keys on the touchscreen dynamically, enlarged the area of the selected characters, and shortened the distance between characters in order to enhance input performance and reduce error rates (Merlin and Raynal, 2010).

Apart from an ergonomic layout matching users' actions, cognitive concepts must be consistent with users' intuition, past experience and memory. Roughly categorized, the methods for text entry are based on two kinds of arrangement: alphabetic order and QWERTY layout. However, which of the two layouts is most suitable for users? Sandnes and Aubert (2007) argued that the QWERTY layout is much easier for users to become familiar with, and therefore is more suitable for the arrangement of letters. Hence, they took the advantage of users' spatial familiarity with the QWERTY layout, and designed a joystick that enabled users to perform text entry. Hwang and Lee (2005) also presented their assertion that the alphabetic order makes it more difficult for user to memorize and results in low typing efficiency. Accordingly, they developed the 3×4 mobile keyboard, the layout of which is similar to QWERTY, resulting in better performance than the standard keyboard with an alphabetic layout.

The cognitive compatibility of encoding is also a very important factor. For instance, Matias et al. (1993) developed the Half-QWERTY chording keyboard, which is operated only with the left hand. The corresponding characters originally typed with the right hand are in the form of mirrored entities. Nevertheless, they did not account for why the keyboard design was based on the mirror approach instead of spatial congruence. Munhall and Ostry (1983) reported that the motional control of the left and right hands is through mirroring rather than direct spatial congruence (cite from Billinghurst et al., 2009). Further, according to the findings regarding the stimulus-response compatibility effect (SRC effect), different hand postures will affect encoding patterns, which will vary with the relationship between visual stimulus and finger reaction (Ehrenstein et al., 1989).

2. Observation: users' typing postures on different phones

Understanding the typing postures of how users operate their mobile phones is crucial for developing new ways of interaction for text entry chording methods. Based on the observation by Gold et al. (2012), three postures were found to occur most often by mobile phone users: holding the phone in one hand while using the index finger of another hand, holding the phone in the palm of the hand while using the thumb of the same hand, and using both hands to hold the phone and typing with both thumbs (Fig. 1). Furthermore, this study tried to analyze the relationship between the typing postures and phones with different typing methods. Therefore, in this stage, the main purpose was to observe the postures when using phones with different typing methods.

Forty-six participants (24 men and 22 women), aged between 18 and 39 ($M = 25.48$, $SD = 4.88$) were recruited from National Cheng Kung University (NCKU). They were asked to perform text entry on their own mobile phone. The results show that mobile phones could be roughly characterized into two types: physical and virtual (touchscreen) keyboards, and the layout of those two keyboards can also be divided into MultiTap and QWERTY. The participants also demonstrated three operating postures that resembled to the result of Gold et al. As Table 1 shown, mobile phones with a physical keyboard and MultiTap style were owned by most participants, and they tended to hold the phone with their right hand and type with their right thumb. Phones with virtual keyboards most frequently featured the QWERTY layout, where participants tended to use the left hand to hold the phone and operate it with their right index finger.

However, to determine which of the above three operating postures is suitable for the new chording methods can be compared via the two perspectives. First, the operating posture needs to avoid the occlusion problem (fat finger problem) (Wigdor et al., 2007) where the user's finger is too big and induces the error of touching neighboring keys. Consequently, some researchers have compared different finger precision of target selection and found that the performance of the index finger was relatively better than the thumb (Sheik-Nainar, 2010; Wang and Ren, 2009). Secondly, the movement time of different operating postures can also be considered as the reason for new design. Silfverberg et al. (2000) compared right hand holding, right thumb typing and left hand holding, right index typing on a MultiTap mobile phone, and used Fitts' law to predict the movement times of two operating postures. They found the input by left hand holding, right index typing achieved a shorter movement time and lowered the index of difficulty. Furthermore, if users held the device with their left hand and typed with their right index finger, the new design could be operated by both hands. This method was found to also reduce the

Fig. 1. Three main text entry postures.

muscles' loading of repetitive operation by a single finger. Therefore, holding the phone in one hand and using the index finger of the other was chosen as the basic posture to develop the new chording text entry methods in this study.

3. The design concept of the new chording text entry methods

Based on the results of above-mentioned studies, this study was consistent with Fitts' law of movement and accuracy. The place where one's hand holds the cellphone was taken as a reference point, and some physical keys were installed on its side. Also, the QWERTY layout was adopted as the foundation of the chording text entry. Naturally, compatibility between the keys and keyboard encoding must be taken into consideration. After observing and comparing how users performed text entry on their cellphones, the design was based on holding with the left hand, and the character selected with the right index finger. Therefore, the authors made text entry possible by combining the physical keys on the side of the cellphone and the virtual keys on the screen. Furthermore, in accordance with Wigdor et al., chording text entry was mainly composed of area-selecting and character-selecting (Wigdor and Balakrishnan, 2003; Wigdor, 2004). Thus, the authors' design of the keys involved two modes of text entry, as described in the following paragraphs. In the first mode, the physical keys are used to select an area, and an interface is then used to input exact characters. In the second mode, the virtual keys are used for area-selecting, and the physical keys are then used to input exact characters.

3.1. MagArea text entry

Under the MagArea text entry method, three keys on the side of a cellphone are used for area-selecting, and the touch interface is used to select exact characters. The three keys on the side respectively represent the upper, middle, and lower rows of the standard QWERTY keyboard. After an area (row) is selected, ten intersected hint boxes appear on the interface, where the characters originally operated with the left or right hand are marked with different colors and placed into different sections. For instance, if character D is the target character, users select the middle row with their left middle finger. Meanwhile, hint boxes with nine characters will appear on the screen, and character D could be selected with their right index finger. As the method is based on chording text entry, users can select an area and exact character simultaneously after they are familiar with the location of each character, as elaborated in Fig. 2.

3.2. MemoryTap text entry

Under the MemoryTap text entry method, five keys on the phone's side are used to select exact characters, and the touch interface is used to select an area. To be more specific, the right index finger clicks the touch interface to select one of the six box areas, and the left hand presses one of the five keys for character-selecting. Furthermore, the mirror principle of the Half-QWERTY keyboard (Matias et al., 1993) as well as the encoding rules based on Orthogonal Stimulus-response Compatibility are followed. As a result, as with standard keyboard encoding, the spatial encoding for the left-handed keys of MemoryTap turns the left-to-right characters into the bottom-to-top ones, i.e., left-bottom/right-top. Contrarily, the spatial encoding for the right-handed keys turns the left-to-right characters into the top-to-bottom ones, i.e., left-top/right-bottom (Ehrenstein et al., 1989). Thus, characters originally input with a right finger are operated with its corresponding finger on the left hand. For example, if character D is the target character, the area on the interface where character D exists is selected with the right index finger. Meanwhile, characters G, F, D, S, and A are organized from top to bottom (left-bottom/right-top) on the physical buttons. Thus, with the middle finger pressing the key, character D is entered successfully. Moreover, the keys on the right side are arranged as a result of mirrored typing. For instance, if character K is the target character, the area on the interface where K exists is selected with the right index finger. Then, characters H, J, K, and L are organized from top to bottom (left-top/right-bottom) on the physical buttons. Thus, with the middle finger pressing the correct key, character K is entered successfully. Similarly, after becoming familiar with the layout, a user can perform synchronous text entry, as illustrated in Fig. 3.

4. Evaluation

Evaluation experiments in this study were primarily intended to understand the learning efficiency, input performance, and error rates of the new text entry methods. Sandnes and Aubert (2007) argued that, for a successful device, ease of learning is a rather crucial factor. In the past, all chording text entry methods required long-term practice and testing, a reality that was severely criticized. Consequently, the new methods have emphasized ease of learning, low learning threshold, and good input performance to be achieved in a short time. Unlike a previous longitudinal study (Lyons et al., 2006), this study required the two new methods to be learned and tested within three days, with their results compared repeatedly. In most previous studies, the learning performance of Multi-Tap has been taken as the control group. However, a virtual

Table 1
Observation results of different postures and mobiles phones.

Postures	Keyboard							
	Physical QWERTY		Physical MultiTap		Virtual QWERTY		Virtual MultiTap	
	N	%	*N*	%	*N*	%	*N*	%
Single hand, single thumb	0	0.0	16	34.8	5	10.9	1	2.2
Both hands, single index	0	0.0	0	0.0	13	28.3	3	6.5
Both hands, both thumbs	1	2.2	3	6.5	4	8.7	0	0.0
Total	1	2.2	19	41.3	22	47.9	4	8.7

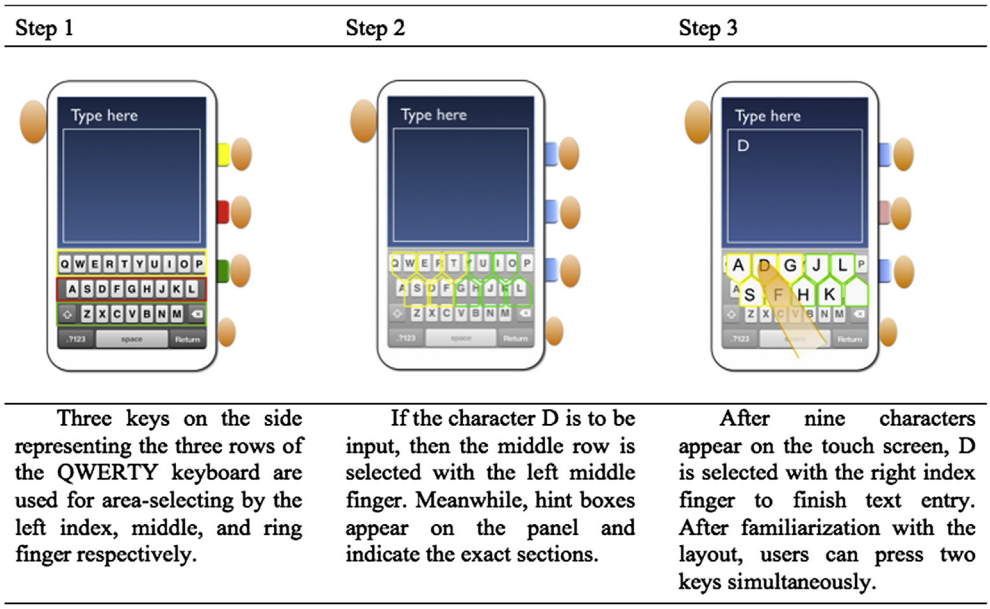

Fig. 2. MagArea text entry.

keyboard is widely-used on touchscreens nowadays. In light of this situation, this study employed the virtual keyboard of an iPhone as the control group. Moreover, it was discovered by other researchers that the learning efficiency of the iPhone virtual keyboard was limited (MacKenzie et al., 2009). For this reason, the experiment participants did not need to spend any time in learning the virtual keyboard. By contrast, they spent three days in learning this study's two new methods. Then, their learning performances of the final tests were compared with the single test result of the iPhone virtual keyboard.

4.1. Participants

The new input methods presented in this study are based on the standard QWERTY keyboard, so participants were limited to right-handers who are familiar with a QWERTY keyboard. According to Ostrach's investigation (1997), average typing speed is 40 words per minute (WPM); therefore, the average typing speed of 40 WPM was taken by this study as a standard to judge whether or not the potential participant was familiar with the QWERTY keyboard. To avoid interference of different dominant hands, participants filled out the Edinburgh Handedness Inventory (Oldfield, 1971) so as to be established them as a right-handed person. Fourteen participants (seven men and seven women) aged between 20 and 28 were recruited from NCKU, with their average age being 22.6. Their range of typing speed on the standard QWERTY keyboard was 41–74 WPM, with an average of 48.86 WPM.

4.2. Equipment and software

The experimental hardware of the two new entry methods included an Apple iPhone 4, on the side of which were self-developed Bluetooth wireless keys. The two kinds of hardware were changed in the same way as a protective cover, as shown in Fig. 4. The cellphone for the MagArea text entry had three physical keys installed on its right side while the cellphone for the MemoryTap text entry had five physical keys. According to the findings with respect to finger positioning (Wu et al., 2011), the keys were properly arranged, with the top right corner of the cellphone regarded as the reference point. In the case of three physical keys, the keys operated with the index, middle, and ring finger were 3.24 cm, 5.61 cm, and 7.67 cm away from the reference point, respectively. As for the five keys, the keys operated with the index finger were 2.74 cm or 3.74 cm away from the reference point, while the keys operated with the middle, ring, and little finger were 5.61 cm, 7.67 cm, and 9.49 cm away from the reference point, respectively. It was confirmed by Wu's findings that the finger positioning mentioned above was not influenced by gender or palm size as far as most users were concerned.

The program interface was written by Apple iOS, which is an application development platform, with the help of the program language Xcode 4.0.1. The finished program was capable of timing, measuring typing speed (WPM), measuring error rates, and storing many articles that could be randomly selected. As shown in Fig. 5, the input interface for the two text entry methods was mainly divided into the timing zone on the top, instant display zone,

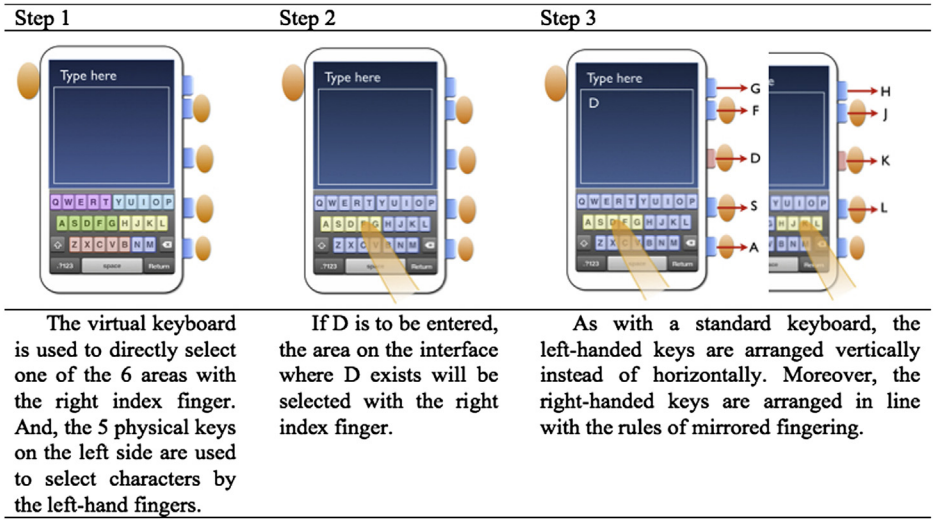

Fig. 3. MemoryTap text entry.

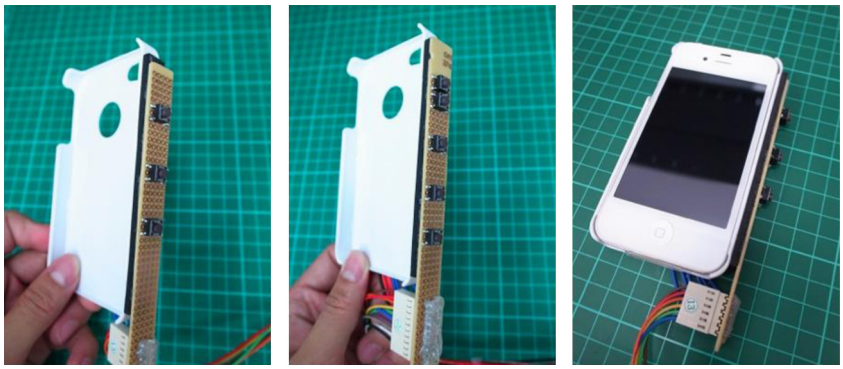

Fig. 4. Experimental equipment for the new text entry methods: the module with thee keys, the module with five keys, and the finished equipment.

article-inputting zone, chording operation zone, and basic function zone. As for article-inputting, when the input text was correct, the character in the instant display zone would turn green, and the user could go on to enter the next character. On the other hand, if the input text was wrong, the character in the instant display zone would turn red. In such a case, the user had to correct the wrong character before being allowed to move on to another. In the case of the MagArea text entry, the interface was composed of ten intersected pentagons, each of which measured about 58*58 pixels (about 10 mm). The letters on the left keyboard were marked in yellow while those on the right were marked in green. As for the MemoryTap text entry, the keys were divided into six groups, and the size of each group varied with the number of letters. The group with five letters measured 170*50 pixels (about 26*5 mm), the group with four letters measured 130*50 pixels (about 20*5 mm), and the group with two letters measured 65*50 pixels (about 10*5 mm). When a user clicked on a certain group, it provided visual feedback by turning from blue to yellow.

4.3. Experimental design

A two factors within-participant design was applied as the experimental design. The two independent variables were the two text entry methods and three training days. Meanwhile, dependent variables were the text entry rate and error rate. The text entry rate (WPM) defined a word as having five characters, whether the character referred to a letter or punctuation, or even empty blank (O'Brien, 2006). Calculation of the error rate was done by extracting the incorrect characters entered in the experiment and presenting them in the form of a percentage of total characters. Suhm et al. (1999) indicated that the four basic factors responsible for errors were substitution, omission, insertion, and transposition. Nevertheless, this study did not focus on the forming mode of errors. Reasonably enough, each participant was asked to correct any error before he/she could continue a trial. The error rate was measured by dividing the incorrect strokes with the total strokes, yielding a percentage.

4.4. Procedure

The experiment was a within-subjects design; therefore, all participants had to learn both new text entry methods and then take tests. The entire experiment lasted 3 days, with participants learning both methods as well as completing tests each day. More specifically, participants learned each of the two methods six times and also received six tests each day. Learning and testing took place alternately, with each trial lasting for 2 min; moreover, there was a two-minute break between two trials. In other words, participants spent about one and a half hours performing the experiment. Additionally, six paragraphs from the famous literary work "Walden" were chosen as the typing material (Liang et al., 2009). Both learning sequences of the text entry methods and typing materials were conducted based on the principle of counterbalance. Therefore, half of the participants were told to learn in one sequence (AB,

Fig. 5. The program interface (from left to right): MagArea text entry, MemoryTap text entry (selected in yellow color), and the screenshot of completed operation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

BA, AB) while the others were told to learn in another sequence (BA, AB, BA) during the three days of the learning process. All participants were told to type the passage shown on the mobile screen as soon and as accurately as possible. Only the final testing trial on each day was recorded as the dependent variables of input performance. As for learning curves, all of the testing trials were recorded as the basis of it. Further, the two new methods were compared with the iPhone virtual keyboard after three days of learning to see which one performed best. After all participants were familiarized with the interface and operation of the iPhone virtual keyboard, they took a single test, with their performances recorded. Subsequently, the input performance of the iPhone virtual keyboard was compared with those of the two new methods during the last testing trial.

5. Results

In this experiment, every participant underwent three days of learning and testing for the two text entry methods. In addition, the virtual QWERTY keyboard of the iPhone was tested and used as the control group. Thus, this experiment was a within-subjects design, and input performances and error rates were all analyzed through repeated measures ANOVA.

5.1. Text entry rate and learning curves

Through repeated measures ANOVA, it was discovered that the text entry rate improved with the increase of days ($F_{2,\ 26} = 177.90$, $p < 0.001$), and that the MemoryTap text entry performed better than the MagArea text entry ($F_{1,\ 13} = 14.49$, $p < 0.01$). There was an interaction effect between the number of days and the two methods, judging from text entry rates ($F_{2,\ 26} = 8.98$, $p < 0.001$). With the least significance difference (LSD) applied to make a comparison, it was revealed that there was no significant difference between the two entry methods on the first day ($F_{1,\ 13} = 0.89$, $p > 0.05$). However, the MemoryTap text entry performed significantly better than the MagArea text entry on the second ($F_{1,\ 13} = 23.05$, $p < 0.001$) and third days ($F_{1,\ 13} = 21.28$, $p < 0.001$). These results indicate that the overall entry rate improved with the increased days and reached the significance level. Also, the interaction was caused by the higher learning efficiency of MemoryTap.

After participants had practiced the two new methods for three days and received the final tests, their input performances were compared with those of the iPhone virtual keyboard. Results show that the MemoryTap text entry had the best performance (mean = 18.78 WPM), followed by the virtual keyboard (mean = 18.33 WPM), while the MagArea text entry had the worst performance (mean = 15.08 WPM). Analyzed through repeated measures ANOVA, results indicate that there were significant differences between the three methods ($F_{2,\ 26} = 10.39$, $p < 0.001$). With LSD applied, it was discovered that the MemoryTap text entry performed significantly better than the MagArea text entry ($F_{1,\ 13} = 21.28$, $p < 0.001$). Meanwhile, the virtual keyboard performed significantly better than the MagArea text entry ($F_{1,\ 13} = 13.77$, $p < 0.01$). However, there was no significant difference between the MemoryTap text entry and the virtual keyboard ($F_{1,\ 13} = 0.22$, $p > 0.05$).

After understanding the differences in learning efficiency of the two entry methods, the authors spread three days of learning results (18 trials and 72 min in total) expand. Further, through the exponential regression function, their learning curve formulas could be obtained. Because there was no learning assignment for the iPhone virtual keyboard, its measured result was a horizontal baseline. As shown in Fig. 6, the learning curve formulas of two new entry methods were obtained, with the formula constants representing their starting points.

From Fig. 6, it can be seen that the MagArea text entry initially had better performance. The exponent in each formula represents the learning efficiency of the text entry method. Moreover, the value x in the figure means a 2 min test after 2 min practice, the

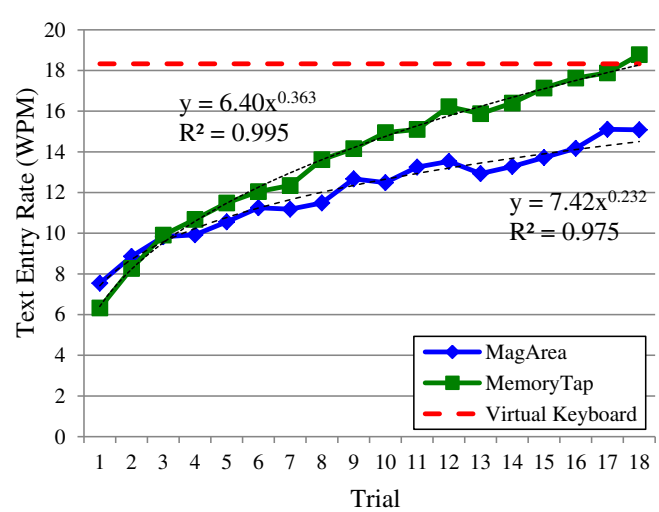

Fig. 6. Detailed trend of three learning performance in text entry rates.

value y represents the predicted text entry performance (WPM) after practice and testing, and the R^2 value means that the curve matched over 97% of the recorded data. The figure shows that, on the first day, there was no obvious difference in learning performance. Then, on subsequent days, the growth rate of the MemoryTap text entry significantly increased. Ultimately, after three days of learning, the entry efficiency of the MemoryTap text entry exceeded that of the virtual keyboard.

5.2. Error rate

During the three days of learning, the error rates of the two entry methods were analyzed through repeated measures ANOVA. It was discovered that their error rates did not reduce with the increase of days ($F_{2,\ 26} = 1.20$, $p > 0.05$). After being compared, the MagArea text entry had a significantly lower error rate than the MemoryTap text entry ($F_{1,\ 13} = 15.57$, $p < 0.01$). The interaction between the number of days and the two methods did not reach a significant level ($F_{2,\ 26} = 2.57$, $p > 0.05$). Thus, during the three days, a stable error rate was found in the two methods. After three days of learning, the last test results of the two methods were compared with that of the iPhone virtual keyboard. As indicated by the statistics, the MagArea text entry had the lowest error rate (mean = 3.91%), followed by the MemoryTap text entry (mean = 5.64%), while the virtual keyboard performed the worst (mean = 6.84%). It was discovered through repeated-measures ANOVA that there was a significant difference between the three methods ($F_{2,\ 26} = 6.23$, $p < 0.01$). With the LSD post-hoc test applied, it was revealed that the MagArea text entry performed significantly better than the MemoryTap text entry ($F_{1,\ 13} = 7.37$, $p < 0.05$) and the virtual keyboard ($F_{1,\ 13} = 9.21$, $p < 0.05$). However, there was no significant difference between the MemoryTap text entry and the virtual keyboard ($F_{1,\ 13} = 1.95$, $p > 0.05$).

Similarly, the participants' learning performances during the three days are represented in greater detail, and reveal how the error rates changed over time. Then, in terms of error rate, the two new methods were compared with the iPhone virtual keyboard. As shown in the above mentioned statistics, which were analyzed through repeated measures ANOVA, the error rates did not reduce with the increase of days, so no learning curve formula was calculated. In the detailed figure of the learning performances below (Fig. 7), there was a trend similar to that suggested by the above statistics. To be more specific, the MagArea text entry had the lowest error rate, and its performance remained rather stable throughout the three days of practice. By contrast, the MemoryTap text entry had a rather high error rate in the first few tests. However, after participants were familiarized with the designated assignments, the MemoryTap text entry performed as well as, and even better than, the virtual keyboard.

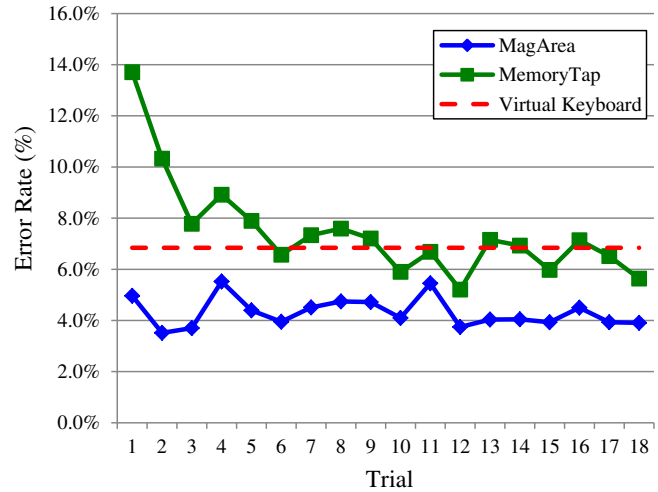

Fig. 7. Detailed trends of learning error rates of the three methods.

6. Discussion

After learning for three days, the two new methods were compared to identify the differences in input efficiency, learning efficiency, and error rates. Under the MagArea text entry, three physical keys were used to select an area, and the graphic user interface was used to select characters. As the physical keys divided the characters into three groups, the area covered by the touchscreen keys was expanded, and thus the error rate of input characters was effectively reduced. Despite this, the distance between the virtual keys failed to be shortened, which slowed input speed. In addition, many participants claimed that the characters, which originally existed in the same row and were entered with two hands, were arranged separately in the upper and lower rows. As a result, operational procedures often occurred in the following manner: the participant first pressed the side key to select a group and then relied on visual search to locate the desired character. Under such a situation, it was very difficult for the participant to input simultaneously with both hands, leading to a lower input rate. But, in terms of error rate, the MagArea method performed better than the existing virtual keyboard; moreover, its input performance can be enhanced in the future by redesigning the physical keys or graphic user interface.

By contrast, the MemoryTap text entry method, in which a graphic user interface was employed to select an area and the physical key was used to select a character, performed better in terms of typing speed. Also, as indicated by the learning curve, the MemoryTap method displayed higher learning efficiency. This is attributed to the encoding of the fingers according with the rules of spatial transpositional correspondence, and the resultant typing habit was very similar to that associated with the standard QWERTY keyboard. As a result, when taking performance tests, all participants did better. However, in the initial process of learning, some participants claimed that they found the MemoryTap more difficult to understand than the MagArea. Yet, after getting used to the way that right-handed characters were mirrored with the left hand, the participants subjectively thought that they had achieved good performance. Meanwhile, they were highly confident with, as well as strongly interested in, the MemoryTap method. Several of the participants even mentioned that implementing a new entry method was just like playing games. Such an attitude helped to strengthen motivation and to enhance input performance, which was reflected in their learning performances and subjective reports.

Furthermore, previous research about chording entry devices has adopted traditional MultiTap as the control group device when input performances were compared. Yet, the ecological significance of such a control group is now no longer adequate. Therefore, the iPhone virtual keyboard was used as the device of the control group and compared with the two new methods. The measured results of the virtual keyboard in our experiment, namely, a mean text entry rate of 18.49 wpm and a mean error rate of 7.7%, showed no obvious difference from the results obtained by MacKenzie et al. (2009). After the three methods were compared, it was discovered that MemoryTap, which took only 72 min of learning, performed as well as the virtual keyboard in terms of typing speed and error rate. By contrast, MagArea performed the worst in terms of typing speed but performed the best in terms of error rate.

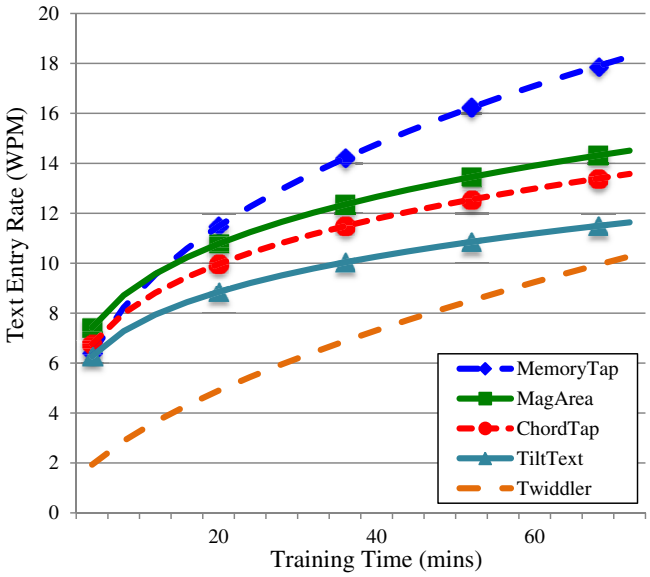

Fig. 8. Learning curves of different chording methods.

Also, by referring to relevant literature, the authors compared the learning performances of different chording text entry devices, as shown in Fig. 8. Subsequently, the same length of learning time was put into the regression formulas for those devices to obtain their estimated performances. To be exact, the learning curve formula for the MemoryTap text entry was $y = 6.40x^{0.363}$, for the MagArea text entry it was $y = 7.42x^{0.232}$, for ChordTap, which is applied to the flip cellphone with three keys on its back, it was $y = 8.41x^{0.241}$ (Wigdor, 2004), for TiltText, which depends on the tilted directions of the cellphone, $y = 7.68x^{0.213}$ (Wigdor and Balakrishnan, 2003), and for Twiddler, which is operated with a single hand, $y = 4.90x^{0.578}$ (Lyons et al., 2006). From the above, it can be seen that the chording text entry methods developed by the authors need only a short period of learning time to achieve better input performance than the other chording text entry methods. The most interesting thing is that MagArea has a learning curve similar to that of ChordTap, which also uses three keys to conduct chording text entry. By contrast, Twiddler, which is available in the market, has a lower learning baseline, but a higher learning index. Therefore, based on its learning curve, it is estimated that the user will have to spend 18 hours in learning before he/she can achieve the same input rate as that of MemoryTap.

7. Conclusion

Based on the rules of ergonomic manual operation and cognitive load, the authors have developed two chording text entry methods, namely MagArea text entry and MemoryTap text entry, which combine physical and virtual keys. In addition to ergonomic factors that were considered carefully in the design process, the main contributions include raising input efficiency and reducing error rates, as indicated by the experimental results. In consequence, the two methods will satisfy users' requirements, perform better than their previous counterparts in terms of learning performance, and enhance users' motivation as well as operational performance.

References

Balakrishnan, V., Yeow, P.H.P., 2008. A study of the effect of thumb sizes on mobile phone texting satisfaction. J. Usability Stud. 3 (3), 118–128.
Billinghurst, M., Guiard, Y., Sellen, A., Zhai, S., 2009. Chapter 6: Case Study 2: Chord Keyboards. In: Human Input to Computer Systems: Theories, Techniques and Technology.
Ehrenstein, W.H., Schroeder-Heister, P., Heister, G., 1989. Spatial S-R compatibility with orthogonal stimulus-response relationship. Percep. Psychophys. 45 (3), 215–220.
Fitts, P.M., 1954. The information capacity of the human motor system in controlling the amplitude of movement. J. Exp. Psychol. 47 (6), 381–391.
Gold, J.E., Driban, J.B., Thomas, N., Chakravarty, T., Channell, V., Komaroff, E., 2012. Postures, typing strategies, and gender differences in mobile device usage: an observational study. Appl. Ergon. 43 (2), 408–412.
Hwang, S., Lee, G., 2005. Qwerty-like 3x4 keypad layouts for mobile phone. In: CHI EA "05 CHI "05 Extended Abstracts on Human Factors in Computing Systems. ACM, pp. 1479–1482.
Kristensson, P., 2007. Discrete and Continuous Shape Writing for Text Entry and Control (Ph.D. Dissertation). Linköping University, Sweden.
Liang, H., Hwang, Y., Chang, F., 2009. Effects of input methods on inter-key press intervals during continuous typing. Ergonomics 52 (9), 1153–1161 (Taylor & Francis).
Logan, G.D., 2003. Simon-type effects: chronometric evidence for keypress schemata in typewriting. J. Exp. Psychol. Hum. 29 (4), 741–757.
Lyons, K., Starner, T., Plaisted, D., Fusia, J., Lyons, A., Drew, A., Looney, E.W., 2004. Twiddler typing: one-handed chording text entry for mobile phones. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, pp. 671–678 (Vienna, Austria).
Lyons, K., Starner, T., Gane, B., 2006. Experimental evaluations of the twiddler one-handed chording mobile keyboard. Hum-Comput. Interact. 21 (4), 343–392.
MacKenzie, I.S., 2002. KSPC (Keystrokes per Character) as a characteristic of text entry techniques. In: Mobile HCI '02 Proceedings of the 4th International Symposium on Mobile Hum-comput Interact, pp. 195–210.
MacKenzie, I.S., Lopez, M., Castelluci, S., 2009. Text entry with the Apple iPhone and the Nintendo Wii. In: Proceedings of ACM CHI 2009 Conference on Human Factors in Computing Systems. Presented at the Proceedings of ACM CHI 2009 Conference on Human Factors in Computing Systems.
MacKenzie, I.S., Soukoreff, R.W., 2002. Text entry for mobile computing: models and methods, theory and practice. Hum-comput Interact 17 (2&3), 147–198.
Matias, E., MacKenzie, I.S., Buxton, W., 1993. Half-QWERTY: a one-handed keyboard facilitating skill transfer from QWERTY. In: Proceedings of the INTERACT "93 and CHI "93 Conference on Human Factors in Computing Systems, pp. 88–94.
McLoone, H., Hinckley, K., Cutrell, E., 2003. Bimanual interaction on the Microsoft Office keyboard. In: Human-computer Interaction: INTERACT'03; IFIP TC13 International Conference on Human-computer Interaction. Ios Pr Inc, Zurich, Switzerland, pp. 49–56.
Merlin, B., Raynal, M., 2010. Evaluation of SpreadKey system with motor impaired users. In: Computers Helping People with Special Needs 12th International Conference, ICCHP 2010, Vienna, Austria, July14–16.
Munhall, K.G., Ostry,, D.J., 1983. Mirror-Image movements in typing. In: Cooper, W.E. (Ed.), Cognitive Aspects of Skilled Typewriting. Springer-Verlag, New York, pp. 247–257.
O'Brien, M., 2006. Effects of Shape, Letter Arrangements, and Practice on Text Entry on a Virtual Keyboard (Master Dissertation).
Oldfield, R.C., 1971. The assessment and analysis of handedness: the Edinburgh inventory. Neuropsychologia 9 (1), 97–113.
Ostrach, T.R.. Typing Speed: How Fast Is Average: 4,000 Typing Scores Statistically Analyzed and Interpreted. http://orbitouch.com/wp-content/uploads/2012/03/Average-OrbiTouch-Typing-Speed.pdf.
Sandnes, F., Aubert, A., 2007. Bimanual text entry using game controllers: relying on users' spatial familiarity with QWERTY. Interacting Comput. 19 (2), 140–150.
Sheik-Nainar, M., 2010. Contact location offset to improve small target selection on touchscreens. In: Proceedings of the Human Factors and Ergonomics Society Annual Meeting. Presented at the Proceedings of the Human Factors and Ergonomics Society Annual Meeting, Vol. 54. SAGE Publications, pp. 610–614.
Silfverberg, M., MacKenzie, I.S., Korhonen, P., 2000. Predicting text entry speed on mobile phones. In: CHI. Presented at the CHI. ACM, New York, USA, pp. 9–16.
Suhm, B., Myers, B., Waibel, A., 1999. Model-based and empirical evaluation of multimodal interactive error correction. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems: the CHI Is the Limit. Pittsburgh, Pennsylvania, United States.
Tinwala, H., MacKenzie, I.S., 2009. Eyes-free text entry on a touchscreen phone. In: 2009 IEEE Toronto International Conference – Science and Technology for Humanity (TIC-STH 2009), pp. 83–88.
Wang, F., Ren, X., 2009. Empirical evaluation for finger input properties in multi-touch interaction. In: CHI 2009. Presented at the CHI 2009. ACM, Boston, MA, USA, pp. 1063–1072.
Wigdor, D., Forlines, C., Baudisch, P., Barnwell, J., Shen, C., 2007. Lucid touch: a see-through mobile device. In: UIST '07 Proceedings of the 20th Annual ACM Symposium on User Interface Software and Technology, pp. 269–278.
Wigdor, D.J., 2004. Chording and Tilting for Rapid, Unambiguous Text Entry to Mobile Phones (Ph.D. Dissertation). University of Toronto, pp. 1–73.
Wigdor, D., Balakrishnan, R., 2003. TiltText: using tilt for text input to mobile phones. In: UIST '03 Proceedings of the 16th Annual ACM Symposium on User Interface Software and Technology, vol. 5, pp. 81–90.
Wu, M.L., Huang, Y.C., Wu, F.G., 2011. The spatial familiarity and moving compatibility of mobile phone text entry. In: International Conference on Planning and Design. Presented at the International Conference on Planning and Design, Tainan.